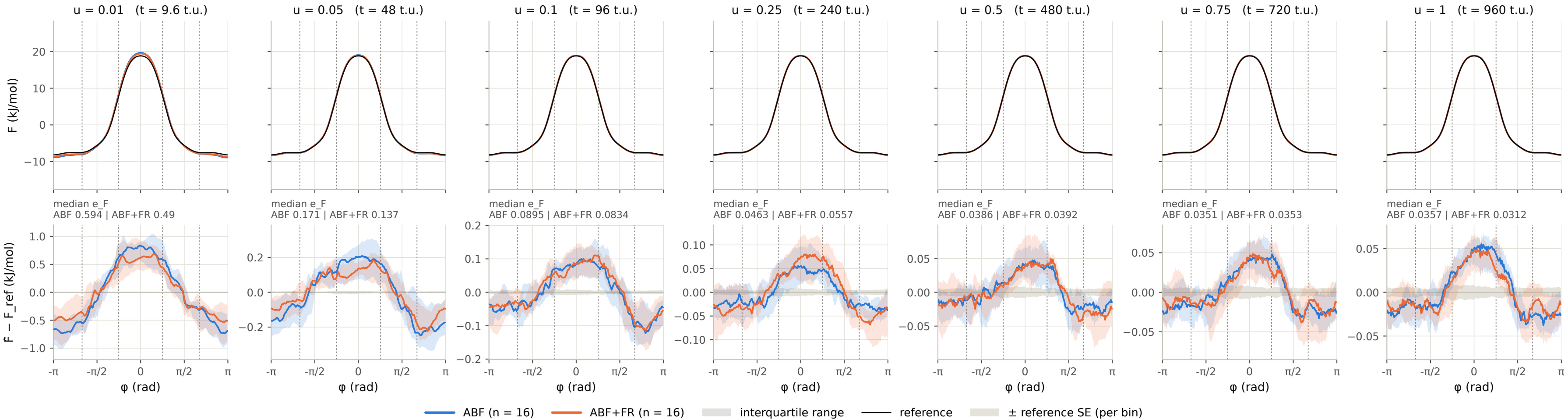


Ethane/LTA 300 K · N = 64 · T_N = 960 t.u.

n_steps = 4,800,000 · h = 0.0002 · B = N n_steps = 3.072e+08 walker-steps · complete seeds: ABF 16/16, ABF+FR 16/16

Free-energy profile snapshots $\bar{F}_t$ vs reference (top) and difference (bottom)



Line = median across seeds; band = interquartile range (25th-75th percentile) across seeds. Metrics are the frozen SCIENTIFIC_PLAN section-4 quantities, unsmoothed.

Profile = histogram_pmf(Gamma), Gamma from the production accumulators once they hold counts; aligned by removing the mean of ($\bar{F}_{\text{hat}} - F_{\text{ref}}$) over the circle: the alignment e_F uses, identical for every method, seed and N. Top: profiles; bottom: difference (y-scale per column).

Dashed guides: $|\phi| = 0.791$ rad (window $|z| < 1.5$ Å) and $|\phi| = 2.109$ rad (cage $|z| > 4$ Å), $z = |\phi| a / 2\pi$, $a = 11.919$ Å. Top-row y range = the reference over the scored window $\pm 35\%$ of its span (larger excursions are clipped there and show in the difference row). No smoothing.

eqb_plot_config/2 · config lta_300K_production.json · 2026-10-10 09:46 UTC